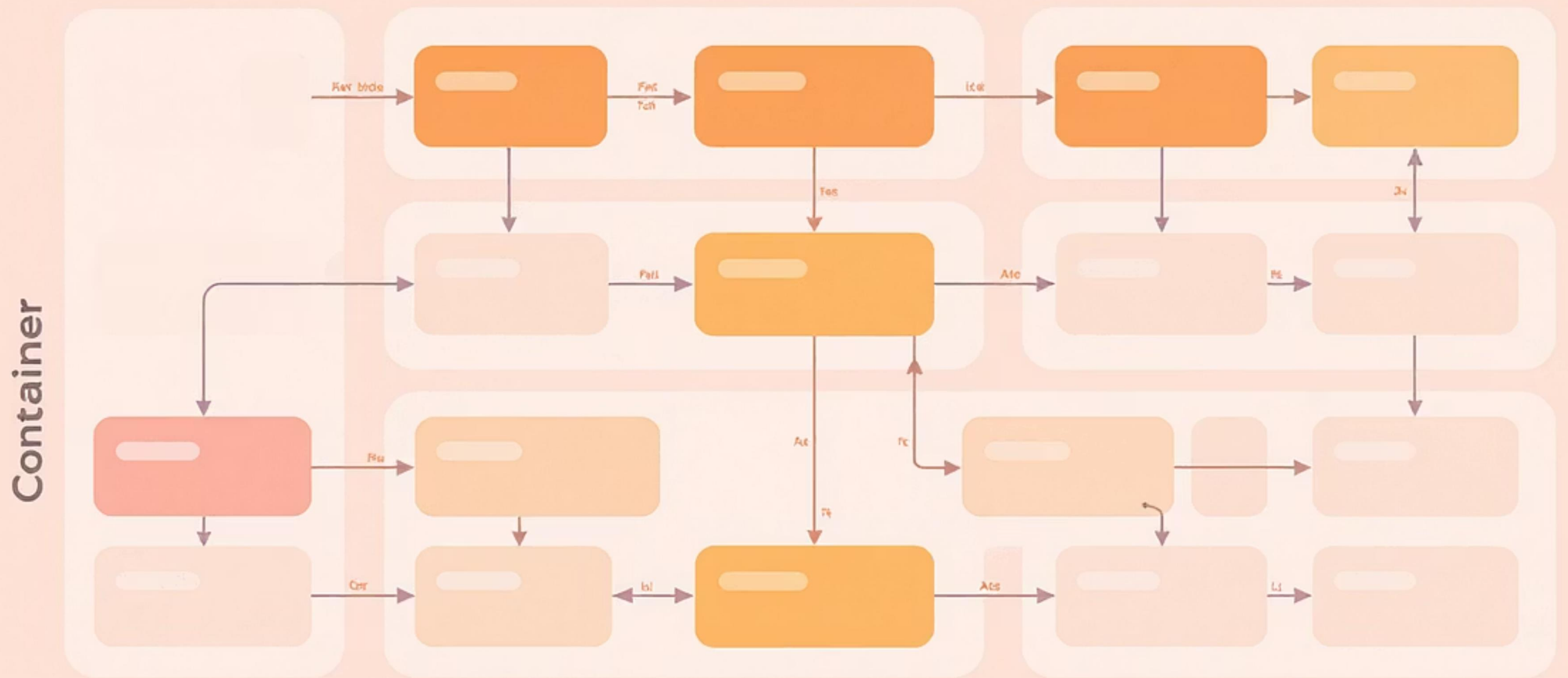




🚢 ☐ Kubernetes: Zero to Hero

Master container orchestration and transform
how you deploy applications

Container Orchestration



What is Kubernetes?

Open-Source Platform

Manages, deploys, and scales containers automatically

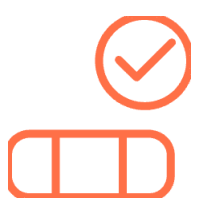
Container Orchestration

Makes hundreds of containers manageable and efficient

Think of Kubernetes as a **smart system administrator** ensuring your apps keep running, even when servers crash.



Why Choose Kubernetes?



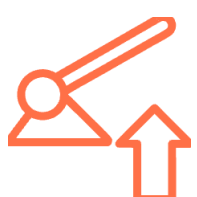
Self-Healing

Automatically restarts crashed containers



Auto-Scaling

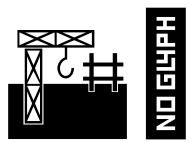
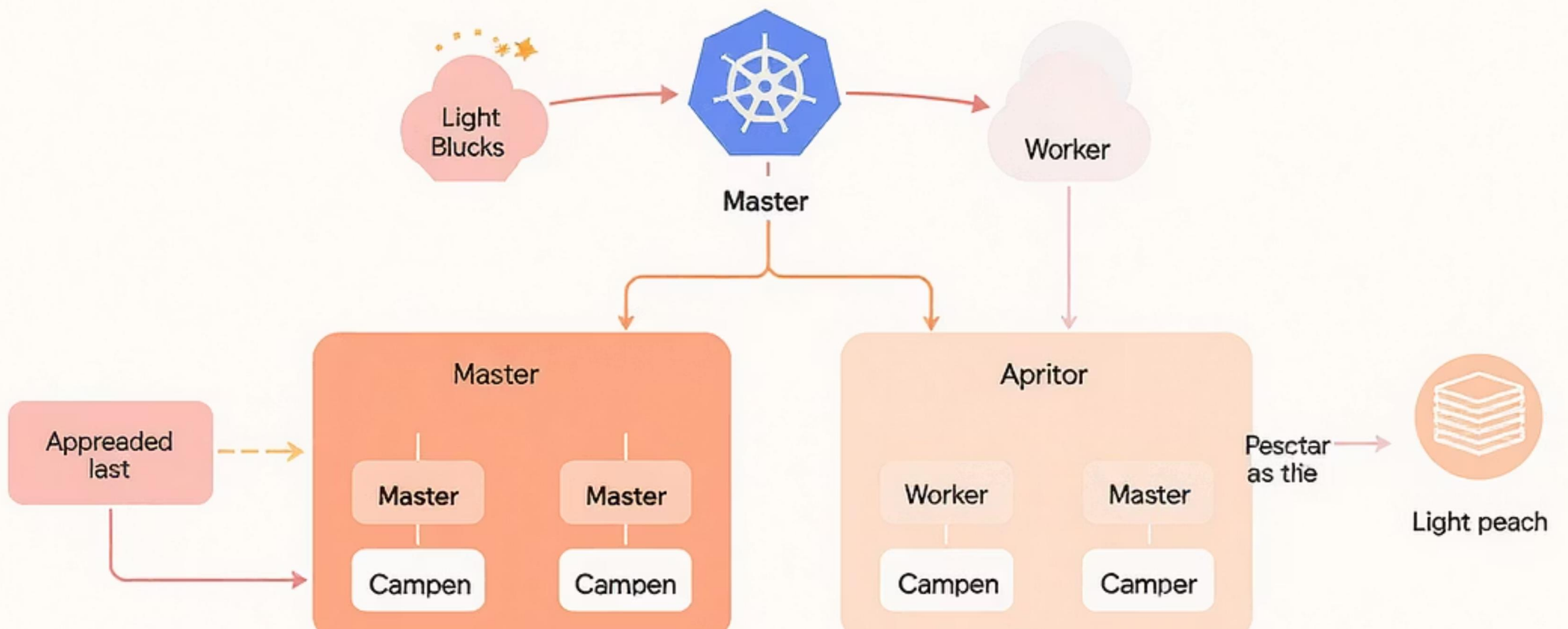
Adds containers when traffic increases



Load Balancing

Distributes traffic evenly across services

Example: During an e-commerce sale, traffic spikes → Kubernetes automatically adds more pods to handle demand.



Kubernetes Architecture



Control Plane

- API Server (the brain)
- etcd (cluster database)
- Scheduler (pod placement)
- Controller Manager (auto healing)

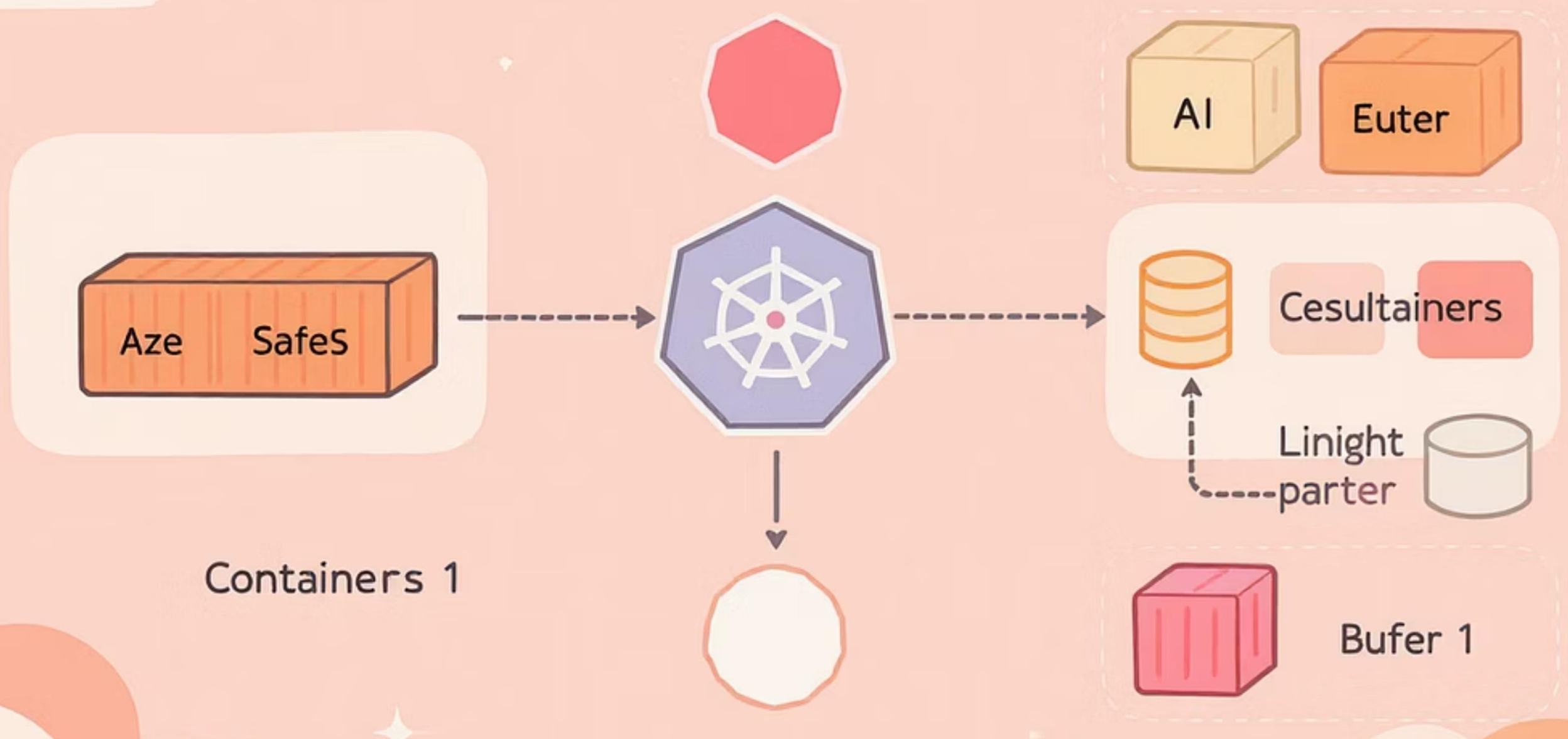


Worker Nodes

- Kubelet (node agent)
- Kube Proxy (networking)
- Container Runtime (Docker/containerd)

Control Plane = Decision Maker | Worker Nodes = Execution Machines

Kubernetes Pod



Understanding Pods

Smallest Unit

The basic deployable unit in Kubernetes

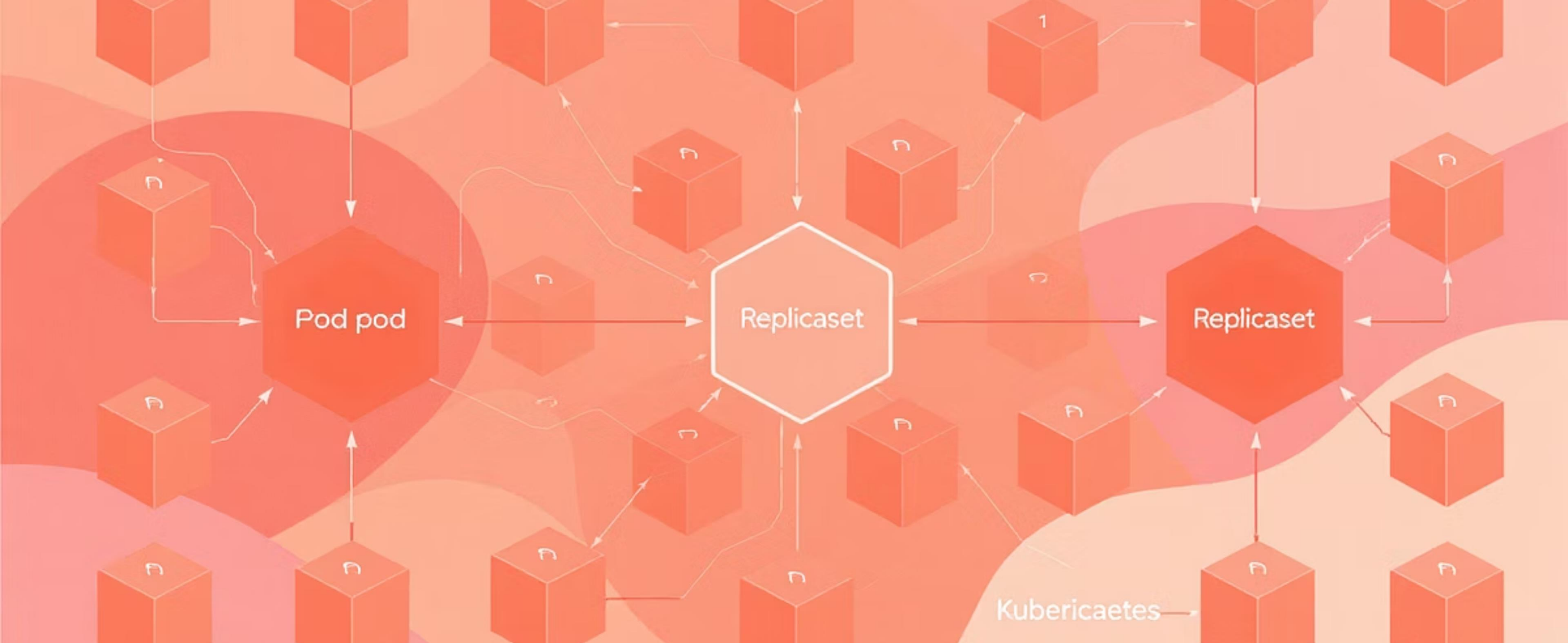
Container Group

Contains one or more containers sharing IP and storage

Auto-Recovery

If a pod dies, Kubernetes recreates it automatically

Example: A web app and its logger container living together in one pod.



↻ ReplicaSets Explained

3

Desired Replicas

Maintains exact number of pod copies

1

Pod Crashes

ReplicaSet creates replacement
instantly

100%

Availability

Ensures continuous service uptime

ReplicaSet ensures your desired number of identical pods are **always running**.



⚙️ Deployments in Action



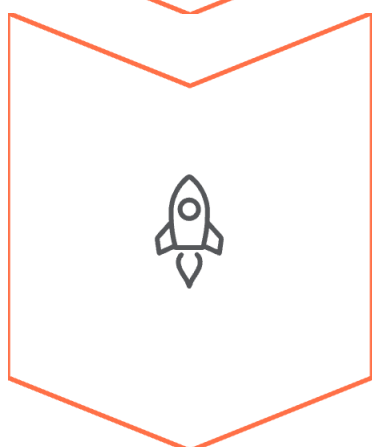
YAML Definition

Describe desired state declaratively



K8s Processing

Handles creation, updates, rollbacks

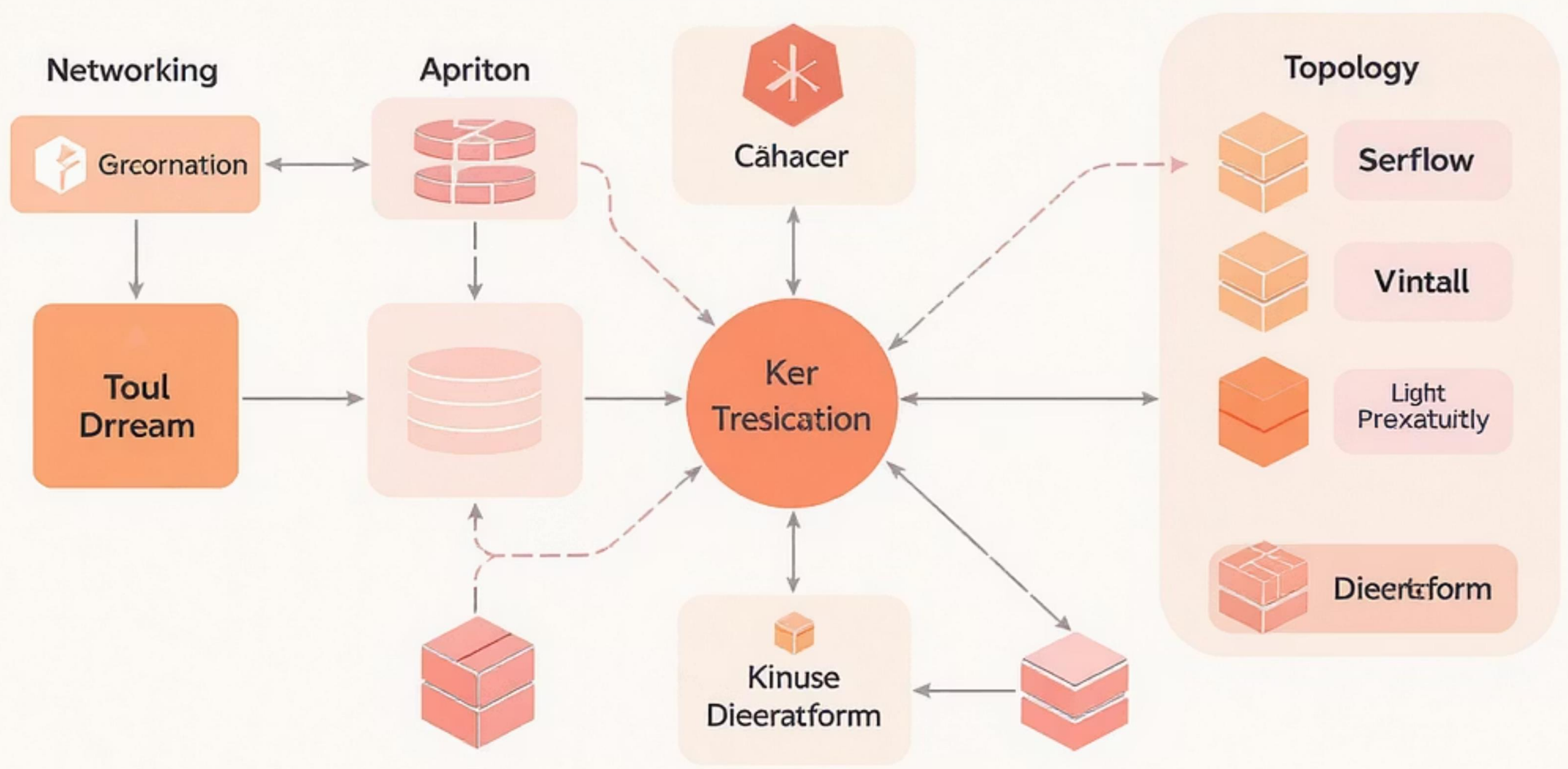


Rolling Updates

Smooth deployment without downtime

Example: Update your app image → Deployment smoothly rolls out new pods.

Kubernetes Service Types



🌐 Services & Networking

ClusterIP

Internal access only (default service type)

NodePort

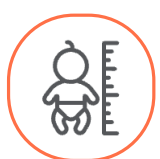
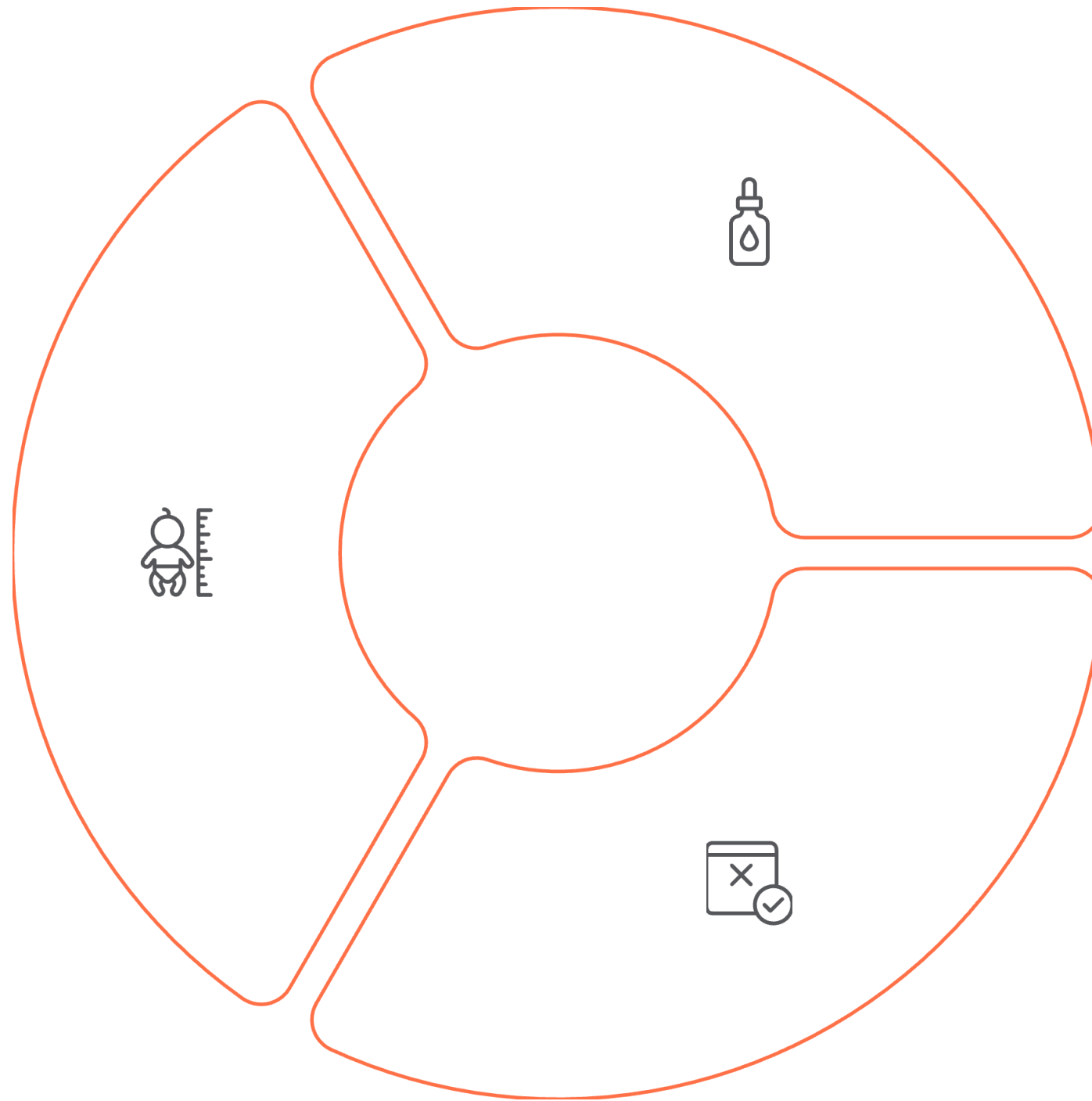
Exposes app via node's IP and fixed port

LoadBalancer

Connects to external world via load balancer

Services provide **stable endpoints** since pod IPs constantly change.

NOGLYPH Namespaces for Organisation



Development

Isolated environment for testing new features



Testing

Quality assurance and staging environment



Production

Live environment serving real users

Divide clusters into **virtual clusters** for different teams and environments.



ConfigMaps & Secrets



ConfigMaps

Store non-sensitive configuration data like URLs and app names



Secrets

Securely store sensitive information like passwords and API keys

Keep environment-specific data **clean and secure** by separating it from your code.

Ready to master Kubernetes? Share this guide with your team! 